

AI-Powered Job Application Assistant

Course Project Proposal

Problem

Job hunting is time-consuming and inefficient. Applicants spend hours manually scanning job postings, tailoring resumes, and writing cover letters — often with little signal on how well their profile actually matches a role. This affects any student or professional actively searching for work.

Data

Two publicly available datasets:

- **LinkedIn Job Postings** (Kaggle, ~500k job listings with titles, descriptions, skills, and requirements)
- **Resume Dataset** (Kaggle, ~2,400 resumes across industries with role categories)

Both are text-based. No custom data collection needed, keeping scope manageable.

Proposed Method

The project has three phases.

Phase 1 — RAG Matching Pipeline Resumes and job descriptions are embedded into a vector database (ChromaDB or Pinecone), and GPT-5-mini is used to retrieve top-matching job candidates for any given resume and generate a relevance score for each pair. This produces a labeled dataset of (resume, job description, match score) triples.

Phase 2 — Fine-Tuning The labeled dataset from Phase 1 is used to fine-tune a smaller open-source model (Llama 3 8B or Mistral 7B) — training it to understand job-fit scoring without needing GPT-5-mini at inference time.

Phase 3 — Personal Assistant Tool The fine-tuned model is deployed as a personal tool. A user uploads their resume, selects a target job posting, and receives:

- A job match score
 - Tailored resume improvement suggestions
 - A draft cover letter
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Expected Results

The primary success metric is whether the fine-tuned model's match scores correlate with GPT-5-mini's scores on held-out pairs (targeting >80% agreement). Qualitatively, success means the personal tool produces actionable resume edits and usable cover letter drafts. A secondary measure is inference speed improvement over GPT-5-mini for the same task.